PHYSICAL CHEMISTRY CUMULATIVE EXAM April 22, 2005

An interesting radioactive reaction is electron capture. In this reaction, a proton in the atomic nucleus reacts with an electron of the atom, changing to a neutron. In the process, the atomic number decreases by one while the atomic mass is approximately unchanged. An example is the radioactive decay of ^{57}Co: $^{57}Co \rightarrow {}^{57}Fe + \gamma$. The gamma radiation emitted is highly monochromatic if the cobalt atom is in the solid state, and is used in Mössbauer spectroscopy. (A less benign use of this reaction has been proposed: the cobalt bomb which would kill all life close to the explosion by its intense gamma radiation but would spare the buildings).

Due to the extremely short range of the nuclear force involved in this reaction (the so-called "weak force"), the reaction takes place only if an electron is within the atomic nucleus. As the wavefunction of *p,d,..* functions is zero at the position of the nucleus, only *s* functions can participate in this reaction, and practically only the 1*s* electrons react. (Note that electrons can pass through nuclei freely; the conventional notion that two things cannot be at the same place simultaneously arises solely from the Pauli principle which prohibits this for *identical* particles.)

(A) Estimate the probability that there is a 1*s* electron within the Co nucleus. The radius of the ^{57}Co nucleus is approximately 4.5×10^{-15} m. The normalized wavefunction of the 1*s* orbital is approximately $\psi(r) = \pi^{-1/2} (Z^*/a_0)^{3/2} \exp[-Z^* r/a_0]$. Here $\exp[x]=e^x$, and $a_0=0.529\times10^{-10}$ m. $Z^* =26.7$ is the *effective* nuclear charge of Co, a little less that the nuclear charge (+27). For the calculation of the probability, the approximate formula

$$\int_0^R r^n e^{-Cr} dr = R^{n+1}(n+1)^{-1} - CR^{n+2}(n+2)^{-1}$$

is useful. It is valid if R is much smaller than C^{-1}. **Do not forget that the volume element for spherical integration is $4\pi r^2 dr$.** (80%)

(B) What are the particles emitted in (a) alpha (b) beta and (c) gamma radiation? (20%)